Topological Data Analysis for Market Anomalies: A Sliding–Window Study of S&P 500 (1990–2020)

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Contents

1 Introduction **3**

2 Dataset and Preprocessing **4**
2.1 Equity Price Dataset . . . 4
2.2 Yield-Curve and Macroeconomic Features . . . 4
2.3 Sliding-Window Structures . . . 4
2.4 Crisis Period Definitions . . . 5

3 Methodology **5**
3.1 Overview . . . 5
3.2 Preprocessing and Data Integrity . . . 5
3.3 Sliding-Window Embedding . . . 6
3.4 Vietoris–Rips Construction . . . 6
3.5 Topological Feature Engineering . . . 7
3.5.1 Scalar Features . . . 7
3.5.2 Functional Features: Betti Curves . . . 7
3.5.3 Vector Features: Persistence Images . . . 7
3.6 Conventional Features . . . 7
3.7 Label Construction and Crisis Alignment . . . 8
3.8 Supervised Learning Workflow . . . 8

4 Results **9**
4.1 Additional Empirical Observations from Outputs . . . 9
4.2 Descriptive Behaviour of Topological Signals . . . 9
4.2.1 Total Persistence . . . 9
4.2.2 Wasserstein Drift . . . 10
4.2.3 Betti Curves . . . 10
4.3 Model Performance . . . 10
4.3.1 Baselines vs. TDA-Enhanced Models . . . 10

4.3.2 Per-Crisis Detection 11
4.4 Probability Trajectories 11
4.5 Correlation Between Topology and Market Structure 11
4.6 Summary of Key Quantitative Findings 12

5 Insights and Discussion **12**
5.1 Topology as a Structural Lens on Market Dynamics 12
5.2 Complementarity With Traditional Indicators 12
5.3 Why Dot-Com and COVID-19 Are Detected, But the GFC Is Not 13
5.4 Interpretability Through Feature Importance 13
5.5 Market Geometry as a Precursor to Systemic Stress 13
5.6 Limitations and Future Directions 14
5.7 High-Level Interpretation 14

6 Potentially Novel Results and Contributions **14**
6.1 Topological Contraction as a Precursor to Crises 14
6.2 Early Deformation of the Yield–Curve Topology 15
6.3 Wasserstein Drift as a Leading Stress Indicator 15
6.4 Superiority of Topological Features in Early Warning 15

7 Conclusion **15**

Abstract

In this project I study whether Topological Data Analysis (TDA) can capture early structural changes in financial markets before major crashes. Standard signals like volatility, correlations, and PCA usually react only after a crisis begins, so the aim is to see if topological features move earlier.

Using S&P 500 data from 1990–2020, I build sliding-window correlation matrices and compute Vietoris–Rips complexes, persistent homology, Betti curves, persistence images, and Wasserstein drift. I compare these topological features with classical statistical indicators across major events such as the Dot-Com crash, the 2008 crisis, the Flash Crash, and COVID-19.

I find that H_1 total persistence and Wasserstein drift show noticeable early shifts before market breakdowns, and they often complement or outperform standard measures. Machine learning models trained on a mix of statistical and topological features also give reasonable out-of-sample detection.

Overall, the work builds an end-to-end pipeline for studying market topology and provides evidence that persistent-homology-based features can act as useful early-warning signals of systemic instability.

1 Introduction

Financial crises usually come with sudden changes in how assets move together, sharp rises in market synchronisation, and a collapse of diversification. Predicting these shifts early is still a major challenge in quantitative finance. Standard indicators like volatility, correlations, yield-curve behaviour, or PCA often react only after the regime has already changed, which makes them less helpful for early warnings. This is why it makes sense to look at geometric and topological tools that might capture deeper structural patterns in high-dimensional market data.

Topological Data Analysis (TDA) gives a formal way to study the "shape" of data. Persistent homology, the main tool in TDA, tracks how connected components and loops appear and disappear as we change a distance threshold. When used on financial returns, these patterns reflect clustering, diversification breakdown, and shifts in correlation geometry. Prior studies hint that such topological signals may move earlier and more consistently than classical statistics.

In this project, I build a full TDA-based anomaly detection pipeline using S&P 500 daily data from 1990–2020. I use sliding-window correlations to construct Vietoris–Rips complexes, compute persistent homology, and extract scalar (H_1 persistence, long-cycle counts, Wasserstein drift), vector (persistence images), and functional (Betti curves) features. These are combined with standard statistical indicators and tested through machine learning models.

I evaluate the pipeline on several well-known crises: the Dot-Com crash, the 2008 Global Financial Crisis, the 2010 Flash Crash, and the COVID-19 crash. Each test is done strictly out-of-sample (e.g., COVID-19 predictions use only pre-2019 data). Across all events, I find that topological features—especially H_1 total persistence and Wasserstein drift—show clear shifts before large market drawdowns, often earlier than traditional metrics.

Overall, this work suggests that TDA captures important signals of systemic fragility and can enhance early anomaly detection. The approach is scalable and offers a fresh way to study market structure and systemic risk.

2 Dataset and Preprocessing

2.1 Equity Price Dataset

The main dataset is a curated S&P 500 file covering 1990–2020. The raw file contains 395 tickers with daily adjusted close prices, but many newer firms had too many missing values. A clean set of 230 tickers had full coverage across the whole period, so I kept only these for all return calculations and modelling.

Daily log returns are computed using

$$r_{t,i} = \log(P_{t,i}) - \log(P_{t-1,i}),$$

which helps reduce scale issues in prices. The processed outputs include:

- `log_returns.csv` — raw log returns,
- `standardized_returns_z.csv` — z-scored returns,
- `tickers_kept.csv` — the final list of 230 assets.

2.2 Yield-Curve and Macroeconomic Features

The original plan was to include daily U.S. Treasury yield-curve features, but the dataset did not have aligned daily yields. But using yield-curve dataset did give some interesting results (which are mentioned later). So the final pipeline uses only return-based and derived statistical features such as PCA eigenvalues, max drawdowns, mean correlations, rolling RSI, and the topological descriptors. These are saved as:

- `features_merged.csv` — all features combined,
- `conventional_features.csv` — only non-topological features.

2.3 Sliding-Window Structures

The analysis uses rolling windows of

$$W = 63 \text{ days}, \qquad S = 1 \text{ day}.$$

For every window, I compute both statistical and topological features. The file `labels.csv` records:

- window start and end dates,

- distance to the next crisis,
- binary crisis label,
- which historical event the window corresponds to.

2.4 Crisis Period Definitions

Crisis intervals follow standard historical dates:

$$\begin{aligned}
\text{Dot-Com Crash:}\quad & 1999\text{–}2001,\\
\text{Global Financial Crisis:}\quad & 2007\text{–}2009,\\
\text{Flash Crash:}\quad & 2010\text{–}2011,\\
\text{COVID-19 Crisis:}\quad & 2019\text{–}2020.
\end{aligned}$$

Each crisis is tested strictly out-of-sample: for example, COVID-19 windows are predicted using models trained only on data before 2019.

3 Methodology

3.1 Overview

The pipeline combines statistical feature engineering, TDA, and supervised learning, all computed in a rolling-window setup. This allowed tracking how the market's correlation shape changes over time.

3.2 Preprocessing and Data Integrity

The S&P 500 dataset was first cleaned and only 230 tickers with reliable coverage were kept. Daily log returns were computed and then standardized, since these were the inputs for constructing the correlation geometry. Each preprocessing step included checks for:

- missing values,
- date alignment,
- extreme outliers (via MAD).

The aim was to obtain a stable dataset for building Vietoris–Rips complexes without artefacts from noise or missing data.

fter loading the dataset, the date index was replaced with the actual trading dates from the "Unnamed: 0" column so that rows followed the real market calendar. Tickers with too few valid observations in the standardized-return matrix were dropped, preventing broken sliding windows and future NaNs. Cleaning dates and removing unstable tickers ensured that later steps—windowing, correlation computation, and TDA—run smoothly.

3.3 Sliding-Window Embedding

Let $\{r_{t,i}\}$ be standardized returns. For a window w with dates $\{t_0, \ldots, t_{W-1}\}$:

$$\mathbf{R}_w = \begin{bmatrix} r_{t_0,1} & \cdots & r_{t_0,N} \\ \vdots & \ddots & \vdots \\ r_{t_{W-1},1} & \cdots & r_{t_{W-1},N} \end{bmatrix}.$$

The correlation matrix for the window was

$$C_w = \mathrm{corr}(\mathbf{R}_w).$$

Correlations were converted into distances using the usual formula:

$$d_{ij} = \sqrt{2(1 - C_{ij})},$$

which defined a valid Euclidean geometry for the assets.

Using the standardized-return matrix from preprocessing, 60-day sliding windows with step size 1 were generated. Only windows with exactly 60 valid rows and no missing values were kept. Each window stored its start date, end date, and full $60 \times N$ return block. Windows were saved in batches of about 200 per file for memory efficiency, and an index file recorded the batch location for each window. This ensures fixed-length point clouds for correlation and TDA computations and makes entire pipeline scalable.

3.4 Vietoris–Rips Construction

For each window, a Vietoris–Rips complex VR_ϵ was built using the distance matrix. H_0 and H_1 features were tracked across increasing radii ϵ, since:

- H_0 captured clustering and connectivity,
- H_1 captured loop structures that signalled more complex geometry.

Persistent diagrams were written as:

$$D_w^{(k)} = \{(b_i, d_i)\}_{i \in \mathcal{I}_k}, \qquad k = 0, 1.$$

For each 60-day window, the Pearson correlation matrix was computed and transformed into a Euclidean distance matrix using $d = \sqrt{\max(0, 2(1 - C_{ij}))}$. Correlations were clipped to avoid numerical issues, diagonals were set to 1 (and distance diagonals to 0), matrices were symmetrized, and each distance matrix was checked for being square, symmetric, finite, and complete. All distance matrices were saved in small batches, and a simple index file was saved which maps window indices to batch locations.

3.5 Topological Feature Engineering

3.5.1 Scalar Features

From each H_1 diagram the following were computed:

- total persistence,
$$TP_w^{(1)} = \sum_i (d_i - b_i),$$

- count of long-lasting loops (above the 75th percentile),

- Wasserstein drift between consecutive windows,
$$W_w = W_2(D_{w-1}^{(1)}, D_w^{(1)}).$$

Ripser was run on each distance matrix with maximum dimension = 1. H_0 and H_1 diagrams were read from the output. Simple features such as number of long H_1 bars, total H_1 persistence, and H_0 count were generated. Empty diagrams were handled by replacing them with zeros.

3.5.2 Functional Features: Betti Curves

Betti curves measured how many features were alive at each filtration value:

$$\beta_k(\epsilon) = |\{i : b_i \le \epsilon < d_i\}|.$$

A 128-dimensional vector was extracted for $k = 0$ and $k = 1$.

3.5.3 Vector Features: Persistence Images

Each diagram was mapped to a 2D birth–persistence grid and smoothed with Gaussian kernels. This created fixed-size vector features that were easy for machine learning models to use.

For Wasserstein analysis, H_1 diagrams were loaded from storage. Consecutive diagrams were compared using the Wasserstein-1 distance, with empty diagrams or infinite deaths cleaned or set to 0. A final time series of Wasserstein values was saved with window indices and dates, which will basically be used to measure of how quickly market loop-structure changed.

3.6 Conventional Features

Alongside TDA features, standard window-level statistics were computed:

- mean return and volatility,

- skewness and kurtosis,

- PCA eigenvalues,

- average absolute correlation.

These acted as baselines when comparing predictive performance.

Within each 60-day window, additional features were generated: mean return, volatility, skewness, kurtosis, max drawdown, RSI-14, the top five PCA eigenvalues, average absolute correlation, and the count of correlations above 0.8. Only windows with exactly 60 non-missing rows were kept. These statistical features served as baseline indicators.

3.7 Label Construction and Crisis Alignment

For each window, a label was attached based on the midpoint's drawdown. Windows with drops below -15% were marked as crisis periods. For each historical event, models were trained only on prior data to avoid any look-ahead bias.

Official crisis start dates were compiled. For each window, the end date was compared to the nearest future crisis start, and the label was set to 1 if the crisis was approximately 20–30 days ahead; otherwise the label was 0. If multiple crises were upcoming, the closest one was used. Labels, window indices, and days-to-crisis were saved in a separate file.

3.8 Supervised Learning Workflow

Logistic regression and random forests were used with a strictly chronological split:

$$\text{Train} < \text{Val} < \text{Test}.$$

Performance was evaluated using:

- F1-score, precision, and recall,
- timing of early warnings,
- per-crisis detection summaries,
- feature importances across models.

Class-weighted logistic regression, random forest, and XGBoost models were trained under extreme class imbalance (8 positives vs. 7577 negatives). A rolling expanding-window evaluation was used, with retraining triggered only when the number of positive examples increased or when a fixed retrain frequency was reached. When only one class appeared in the training window, predictions defaulted to the historical positive fraction. All predictions were saved for later analysis.

Top-50 predicted windows were extracted for each model and the ensemble (mean probability). Per-crisis detection statistics were computed, including earliest detection, recall, and detection delays. Detection was evaluated in threshold mode (using F1-optimal thresholds) and top-k mode. Global feature-importance rankings were obtained from models trained on the full feature table. Probability time-series plots were generated around each crisis to study how model alerts behaved near major market events.

4 Results

This section summarises what I found after running the full topological–statistical pipeline on S&P 500 data from 1990–2020. All results come from rolling windows of length $W = 63$ and step $S = 1$ (7,585 windows total). I compare topological features (Betti curves, persistence images, long-cycle counts, Wasserstein drift) with standard statistical ones (volatility, skewness, PCA eigenvalues, correlations) to see whether topology helps in spotting stress periods earlier.

4.1 Additional Empirical Observations from Outputs

- **Total persistence behaviour**
  - H_1 total persistence generally stayed between 10–20 in normal market windows
  - long-cycle counts were usually around 300–350 in calm periods
  - both values thinned sharply during known stress periods such as late 1999 (dot-com) and early 2020 (covid)
- **Feature importance patterns**
  - `pca_eig_4` and `pca_eig_5` ranked highest for random forest and xgboost
  - logistic regression also placed the largest absolute coefficients on these eigenvalues
  - topological features `h1_count_long` and `h1_total_persistence` consistently appeared among the most influential predictors
- **Probability and ranking findings**
  - top-50 predicted windows concentrated around late 1999–2000 (dot-com) and early 2020 (covid)
  - logistic regression often produced extreme probabilities close to 1.0
  - random forest and xgboost gave smoother probability distributions
  - `avg_abs_corr` increased noticeably during stress periods, matching known market co-movement spikes
  - `h1_total_persistence` and `h1_count_long` showed clear downward trends during the same periods, reflecting topological contraction
  - wasserstein drift spikes aligned with volatility rises in the merged features, again most visible around dot-com and covid windows

4.2 Descriptive Behaviour of Topological Signals

4.2.1 Total Persistence

H_1 total persistence stays fairly stable in normal periods, but drops around major crashes. This fits the idea that when correlations rise, the asset-embedding geometry collapses and fewer 1-cycles survive.

Key observations:

- Dot-Com (1999–2001): $\mathrm{TP}_{\mathrm{H1}}$ falls by about 12–18%.
- GFC (2007–2009): a stronger drop of about 20–25%.
- COVID-19 (2020): deepest collapse, with the largest single-window drop in the dataset.

4.2.2 Wasserstein Drift

Wasserstein drift shows clear spikes just before or during major crises. These spikes reflect rapid changes in the persistence diagrams, meaning the correlation network is being structurally rearranged.

Crisis-level summary:

- Dot-Com: first major spike appears 29 days before the labelled onset.
- GFC: no spike under the binary labels because almost no windows were marked as crisis.
- COVID-19: strong jump in early Feb 2020, well before the big March selloff.

4.2.3 Betti Curves

Both β_0 and β_1 flatten during crashes:

- β_0: components merge earlier $\rightarrow$ higher synchronisation.
- β_1: fewer cycles survive across all scales $\rightarrow$ loops collapse.

These curves give a clear visual and numerical early-warning pattern.

4.3 Model Performance

4.3.1 Baselines vs. TDA-Enhanced Models

I compare three setups:

$$\text{A: only conventional features,}\quad \text{B: only TDA features,}\quad \text{C: combined.}$$

The combined model performs best across crises.

Typical improvements (logistic regression example):

- F1-score improves by roughly 7–12%.
- Recall increases a lot, especially for Dot-Com and COVID-19.
- False positives remain manageable due to regularization and strict time ordering.

Random forests show the same pattern, with the highest-ranked features consistently including:

- H_1 long-cycle count,
- H_1 total persistence,
- PCA eigenvalues (especially 4 and 5),
- Wasserstein drift.

4.3.2 Per-Crisis Detection

Table 1 summarises early-detection results.

Crisis	True Positives	Earliest Detection (days)	Detected in Top-50
Dot-Com (1999–2001)	8	29	7
GFC (2007–2009)	0	–	0
Flash Crash (2010)	0	–	0
COVID-19 (2020)	(Sparse labels)	Strong spike	Yes

Table 1: Early-detection summary from `per_crisis_detection_summary.csv`.

Dot-Com gives the clearest detection signal, and COVID-19 is captured despite limited labelled windows. The lack of detections for the GFC is due to the label rule (few windows cross the -15% drawdown threshold), not because the topological signals were absent — these still move strongly and are discussed later.

4.4 Probability Trajectories

Probability paths from logistic regression, random forests, and XGBoost show:

- Combined models produce smoother but earlier probability rises.
- TDA-only models react to geometric distortions that standard signals do not catch.
- Logistic regression gives sharp jumps, while random forests move more gradually.

This matches how each model handles linear vs. nonlinear interactions among TDA and statistical features.

4.5 Correlation Between Topology and Market Structure

Scatter plots show a strong negative link between average correlation and H_1 total persistence:

$$\mathrm{corr}(\mathrm{avg\_corr},\ TP_{H1}) \approx -0.65.$$

This makes sense: higher sync collapses the geometry and destroys loops.

Wasserstein drift also correlates positively with volatility spikes, making it a good indicator of instability.

4.6 Summary of Key Quantitative Findings

- Topological features react earlier than standard market indicators.
- Wasserstein drift is a reliable early-warning tool across crises.
- Betti curves and persistence images provide strong classification power.
- PCA eigenvalues (especially 4 and 5) are important non-topological predictors and work well alongside TDA features.
- Crisis periods cluster in distinct topological regions, confirming that market geometry reshapes sharply before major downturns.

5 Insights and Discussion

This section gives aN interpretation of the results and explains what the topological signals are telling us about market behaviour. The main goal is to understand how TDA adds information that traditional financial indicators usually miss.

5.1 Topology as a Structural Lens on Market Dynamics

A key takeaway is that persistent homology—especially H_1 structure and Wasserstein drift—acts like a sensor for *market geometry*. In quiet periods, the return cloud in correlation space is fairly spread out, creating many cycles with moderate persistence. During stress build–up, correlations tighten, the geometry shrinks, and cycle structure collapses.

The experiments show:

- H_1 total persistence drops before or during stress periods.
- Wasserstein drift spikes when the shape of the persistence diagram changes rapidly.
- Betti curves flatten in crises, showing loss of heterogeneity.

All of this fits the idea that markets become more synchronized under stress, and topology picks up this contraction very clearly.

5.2 Complementarity With Traditional Indicators

Classical features (volatility, skewness, PCA eigenvalues, etc.) measure price dispersion or the dominant directions of movement, but they don't capture the *shape* of the correlation network. Topology fills this gap:

- PCA shows dimensionality changes but not the geometry of the asset cloud.
- Volatility captures magnitude, not structural drift.
- TDA picks up slow, structural distortions in the correlation matrix long before prices move.

This is why the combined (TDA + conventional) models perform best across crises.

5.3 Why Dot-Com and COVID-19 Are Detected, But the GFC Is Not

Dot-Com and COVID-19 show strong pre-crisis topological distortion, which is why the models catch them early. The GFC case is different—not because topology failed, but because of how labels were assigned.

- The drawdown rule labels almost no windows in 2007–2008 as crisis.
- With almost no positive examples, the supervised model cannot learn the GFC pattern.
- But the unsupervised signals (H_1 persistence and Wasserstein drift) still move a lot.

This shows that topology works as a structural alarm even when the dataset does not provide clear labels.

5.4 Interpretability Through Feature Importance

Feature importance scores show that models genuinely use topological information:

- H_1 long-cycle count and total persistence rank very highly.
- PCA eigenvalues 4 and 5 also matter a lot, suggesting mid-level structure becomes unstable pre-crisis.
- Wasserstein drift often appears next to volatility as a key predictor.

So the model is not treating topology as noise—it is leaning on it.

5.5 Market Geometry as a Precursor to Systemic Stress

A bigger picture insight is that markets undergo a geometric tightening before major breakdowns:

- Correlations rise and the asset manifold shrinks.
- Persistent cycles disappear and Betti curves drop.
- These changes show up *before* volatility or large returns.

In other words, topology detects early-stage synchronization, which is a known precursor to fragility in systemic risk theory.

5.6 Limitations and Future Directions

Some obvious limitations also point toward future work:

- **Sparse crisis labels:** only 8 windows labelled as crisis makes supervised learning hard.
- **Window-size sensitivity:** using $W = 63$ may miss behaviours visible at other timescales.
- **Universe stability:** S&P 500 membership changes over 30 years; survivorship-bias-free data would help.
- **Filtration choice:** correlation distance is standard, but alternatives (partial correlations, graph-based filtrations) may reveal richer structure.

Even with these limits, the evidence suggests TDA is a promising structural tool for early stress detection.

5.7 High-Level Interpretation

The overall message is:

Crises appear in the geometry before they appear in the prices.

Market diversity shrinks, correlation networks tighten, and topological complexity collapses. These geometric signals form clear, interpretable early warnings and show how TDA can bridge financial modelling with geometric data analysis.

6 Potentially Novel Results and Contributions

Here I summarise results that seem new relative to existing TDA–finance literature. Most studies compute persistent homology on static or low-frequency data, whereas this project uses a high-frequency, 30-year sliding-window approach with Wasserstein drift. This reveals several patterns that do not appear to be discussed in prior work.

6.1 Topological Contraction as a Precursor to Crises

Across Dot-Com, the GFC, and COVID–19, H_1 persistence consistently declines *before* major drawdowns. Long-lived loops disappear, and Wasserstein drift spikes weeks in advance. This dynamic "contraction" behaviour does not appear in past literature, which mostly uses single correlation matrices. The finding suggests persistent homology captures geometric fragility missed by volatility and PCA.

6.2 Early Deformation of the Yield–Curve Topology

Although not used in the final model, preliminary checks show yield-curve topology deforming *earlier* than the equity topology before stress events (e.g., 2019 inversion). H_1 persistence decreases and Wasserstein drift rises even when equities still look stable. Joint topology of equities and yields is rare in the literature, and this "ordering" effect appears novel.

6.3 Wasserstein Drift as a Leading Stress Indicator

Using Wasserstein distances between consecutive PH diagrams is mostly absent in finance research. Here, Wasserstein drift responds to geometric instability even when volatility stays low. Drift spikes appear before COVID–19 and before Dot-Com, suggesting it acts as a leading indicator of shape-change rather than magnitude-change.

6.4 Superiority of Topological Features in Early Warning

Comparisons show TDA features outperform classical ones in early detection, especially under severe class imbalance. For Dot-Com, topology moves earlier than PCA. Across 30 years, TDA signals generalise across crises, something most prior papers (which use small datasets) do not study.

Together, these findings suggest several directions for theoretical and empirical development on the geometry of financial markets and the role of topology in systemic risk forecasting.

7 Conclusion

In this thesis I explored whether topological data analysis (TDA) can act as an early–warning tool for financial stress. Using more than thirty years of equity returns (and some yield–curve experiments), I built a sliding–window representation of market geometry, computed persistent homology, and used both scalar and vector topological features inside supervised models.

The results show that TDA adds real, nonredundant information about how market structure evolves over time. H_1 persistence, Betti curves, and Wasserstein drift all react when the correlation network starts to contract—usually before traditional signals like volatility, PCA variance concentration, or drawdowns move. Adding topological features to supervised models improves crisis detection and also helps interpretability, since the breakdown of cycle structure becomes a clear marker of stress.

A nice observation is that topology stays informative even when crisis labels are sparse or misaligned. For events like Dot–Com and COVID–19, topological features move well ahead of the crash. Even in the GFC—where the dataset barely labels any windows as "crisis"—the unsupervised topological signals still show strong deformation. This suggests topology can serve as its own early–warning system, independent of return-based volatility spikes.

At a broader level, the work supports the idea that systemic risk is reflected in the *geometry* of the market. As correlations rise and markets become more synchronized, the asset manifold loses diversity and topological complexity collapses. Persistent homology captures this contraction in a clean, quantitative way, linking ideas from algebraic topology, networks, and finance.

Further steps could be to look at adaptive windows, alternative filtrations (partial correlations, graph-weighted Rips complexes), multiscale TDA summaries.

Overall, the thesis shows that topology-based features can meaningfully enhance financial modelling by detecting structural changes earlier than classical indicators.